

Comprehensive Machine Learning Project Report

1. Project Objective

The primary goal of this project was to develop a machine learning model capable of accurately classifying data into two categories: "0" (representing failed launches) and "1" (representing successful launches). A key requirement was to create a model that could effectively identify the minority class (failed launches) to minimize false negatives, a metric known as **recall**.

2. Methodology

The project followed a systematic machine learning workflow, which included:

- **Data Preprocessing:** The dataset was loaded and preprocessed to ensure it was suitable for modeling. This included preparing the features (X) and the target variable (y) and ensuring a stratified split for training and testing.
- **Feature Scaling:** A StandardScaler was used to standardize the features. This is a critical step for distance-based algorithms like SVM and KNN and generally improves the performance of many other models.
- **Model Selection and Tuning:** A wide range of classification algorithms were systematically evaluated, with each model being fine-tuned using GridSearchCV to find the optimal hyperparameters. This process involved fitting each model to 5-fold cross-validation on the training data.

3. Model Evaluation

The models were evaluated primarily using a classification report, which provides detailed metrics beyond simple accuracy. The key metrics focused on were:

- **Precision:** The ratio of correctly predicted positive observations to the total predicted positives.
- **Recall (Sensitivity):** The ratio of correctly predicted positive observations to all observations in the actual class. This was the most important metric for the minority class (failed launches).
- **F1-Score:** The weighted average of Precision and Recall.
- **Accuracy:** The proportion of correct predictions out of all predictions.

4. Model Performance and Analysis

Initial Models (Before Random Forest)

Initial testing with several common classification algorithms showed a consistent limitation in

correctly identifying the minority class.

Model	Best Hyperparameters	Class 0 Recall	Class 1 Recall
Logistic Regression	{'C': 100, 'solver': 'liblinear'}	0.50	0.92
Decision Tree	{'max_depth': 10, 'min_samples_leaf': 4, 'criterion': 'gini'}	0.50	0.92
SVM (Scaled)	{'C': 10, 'gamma': 0.001, 'kernel': 'rbf'}	0.50	0.92
XGBoost	{'colsample_bytree': 0.8, 'learning_rate': 0.1, 'max_depth': 5, 'n_estimators': 200}	0.50	0.92
K-Nearest Neighbors	{'metric': 'manhattan', 'n_neighbors': 9, 'weights': 'uniform'}	0.50	1.00

The consistent recall of 0.50 for the failed launches (Class 0) across a range of different models indicated that they were all struggling to distinguish this class from the successful launches.

Optimal Model: Random Forest Classifier

The Random Forest model demonstrated superior performance in addressing the core challenge of the project.

First Run Results:

- **Tuned Hyperparameters:** {'criterion': 'entropy', 'max_depth': None, 'min_samples_leaf': 1, 'n_estimators': 100}
- **Classification Report:**
 - **Class 0 Recall: 0.67**
 - **Class 1 Recall: 0.83**
 - **Accuracy: 0.78**
- **Analysis:** This model was the first to successfully break the 0.50 recall barrier for the minority class, correctly identifying two-thirds of the failed launches.

Expanded Tuning Results:

- **Tuned Hyperparameters:** {'criterion': 'gini', 'max_depth': 10, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'n_estimators': 300}
- **Classification Report:**
 - **Class 0 Recall: 0.67**
 - **Class 1 Recall: 0.83**
 - **Accuracy: 0.78**
- **Analysis:** The expanded GridSearchCV found different optimal parameters, but the

core performance metrics remained unchanged. This suggests that the model's performance on this dataset is stable and that we have likely reached the performance ceiling for the current feature set.

5. Conclusion

Based on a thorough and systematic evaluation of multiple machine learning models, the **Random Forest Classifier** is the optimal choice for this project. While other models may have matched its overall accuracy, none were able to match its ability to correctly identify the minority class. The consistent recall of 0.67 for failed launches makes it a reliable and effective tool for addressing the project's primary objective.

The next steps for this project would involve using this final, best-performing model to make predictions on new, unseen data, and considering strategies to improve the model further, such as collecting more data or engineering new features.